

8

What is the primary purpose of the Test and Set instruction? 

- ☐ To handle I/O requests.
- ☐ To schedule processes.
- ☒ To provide a hardware-based solution for mutual exclusion.
- ☐ To manage memory allocation.

 **Will be reviewed**

9

What is the primary use of binary semaphores? 

- ☐ To schedule processes.
- ☐ To manage memory allocation.
- ☐ To count the number of available resources.
- ☒ To implement mutual exclusion.

 **Will be reviewed**

10

Which scheduling algorithm gives each process a fixed time slice and switches between them in a circular manner? 

- ☒ Round Robin (RR)

- ☐ First-Come, First-Served (FCFS)
- ☐ Shortest Job First
- ☐ Priority Scheduling

 **Will be reviewed**

11

In which mode does the operating system execute system calls? 

- ☐ Protected mode.
- ☒ Kernel mode.
- ☐ Virtual mode.
- ☐ User mode.

 **Will be reviewed**

12

In which mode do user applications typically run? 

- ☐ System Mode.
- ☐ Kernel Mode.
- ☐ Supervisor Mode.
- ☒ User Mode.

 Will be reviewed

13

How does the **DMA** controller **inform** the CPU that the transfer is complete?



- ☐ By sending a signal to the MMU.
- ☐ By writing a flag to the hard drive.
- ☒ By sending a message through the memory bus.
- ☐ **By triggering an interrupt.**

 Will be reviewed

14

What is an interrupt in an operating system? 

- ☒ **A signal that pauses a running process to handle an event.**
- ☐ A signal that terminates a running process.
- ☐ A signal that restarts the operating system.
- ☐ A signal that allocates memory to a process.

 Will be reviewed

15

In the Dining-Philosophers Problem, what do the "**philosophers**" represent? 

- ☐ Shared resources.
- ☐ I/O devices.
- ☒ Processes.
- ☐ Memory blocks.

 **Will be reviewed**

16

In the Dining-Philosophers Problem, what do the "chopsticks" represent? 

- ☒ Shared resources.
- ☐ Process states.
- ☐ Memory addresses.
- ☐ I/O requests.

 **Will be reviewed**

17

What is the main challenge in the Dining-Philosophers Problem? 

- ☒ Preventing deadlock and starvation.
- ☐ Maximizing CPU utilization.
- ☐ Ensuring that philosophers eat simultaneously.

☐ Minimizing memory usage.

 **Will be reviewed**

18

What is the **main constraint** in the **Readers-Writers Problem**? 

- ☐ One reader and one writer can access the resource simultaneously.
- ☒ **Multiple readers can access the resource simultaneously, but writers must have exclusive access.**
- ☐ Multiple writers can access the resource simultaneously.
- ☐ Only one process can access the resource at any time.

 **Will be reviewed**

19

What is the main goal of the critical-section problem? 

- ☒ **To ensure that only one process can execute its critical section at a time.**
- ☐ To simplify memory management.
- ☐ To prevent deadlock in the system.
- ☐ To increase CPU utilization.

 **Will be reviewed**

20

What is a "race condition"? 

- ☒ A condition where the output of a program depends on the unpredictable order of execution of processes.
- ☐ A condition where processes are waiting for I/O operations.
- ☐ A condition where processes are accessing different memory locations.
- ☐ A condition where processes run in parallel.

 **Will be reviewed**

21

In most implementations, what value does the **Test and Set** instruction return?



- ☐ The new value set in memory.
- ☐ The original value read from memory.
- ☒ A boolean indicating success or failure.
- ☐ A random value.

 **Will be reviewed**

22

What is a "spin lock"? 

- ☐ A lock that puts a process to sleep until it is released.

- ☐ A lock that uses semaphores for synchronization.
- ☐ A lock that is only used for I/O operations.
- ☒ A lock that causes a process to repeatedly check a variable until it becomes available.

 **Will be reviewed**

23

What are the two primary operations performed on semaphores? 

- ☐ Start and stop.
- ☐ Allocate and deallocate.
- ☐ Read and write.
- ☒ P (wait/acquire) and V (signal/release).

 **Will be reviewed**

24

What is the primary use of counting semaphores? 

- ☐ To control access to a limited number of resources.
- ☐ To manage interrupts.
- ☐ To manage file systems.
- ☒ To implement mutual exclusion.

 Will be reviewed

25

What is the benefit of using DMA? 

- ☐ Simplified process scheduling.
- ☐ Reduced memory usage.
- ☐ Increased CPU utilization.
- ☒ Improved I/O performance and reduced CPU overhead.

 Will be reviewed

26

How does a user program request a service from the operating system? 

- ☐ By writing data to a specific memory location.
- ☐ By generating a hardware interrupt directly.
- ☐ By directly accessing kernel memory.
- ☒ By executing a system call.

 Will be reviewed

27

What is the major disadvantage of First-Come, First-Served (FCFS) scheduling?



- ☐ It is complex to implement.
- ☐ It has high context switching overhead.
- ☒ It can lead to the convoy effect.
- ☐ It can lead to starvation.

 **Will be reviewed**

28

What is priority inversion in an operating system? 

- ☐ A situation where all processes have the same priority.
- ☒ A situation where a high-priority process is blocked by a low-priority process.
- ☐ A situation where the operating system assigns priorities randomly.
- ☐ A situation where a low-priority process preempts a high-priority process.

 **Will be reviewed**

29

A small computer has 4 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using FIFO algorithm? 

- ☐ 8
- ☐ Others
- ☐ 10

☐ 15☒ 14☐ 12

 **Will be reviewed**

30

A small computer has 4 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using least-recently-used (LRU) algorithm? 

☐ 8☒ Others☐ 10☐ 14☐ 12

 **Will be reviewed**

31

A small computer has 4 page frames. A process makes the following list of page references: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults using optimal page replacement algorithm? 

☒ 8☐ 14

☐ 12☐ 10☐ Others

 **Will be reviewed**

32

In a page memory, the page hit ratio is 0.9. The time required to access a page in secondary memory is equal to 100ns. The time required to access a page in primary memory is 10ns. The average time required to access a page is 

☐ 9 ns☐ 10 ns

$$0.9 * 10 + 0.1 * 100 = 19$$

☐ 90 ns☒ 19 ns

 **Will be reviewed**

33

If there are 32 pages, each of size 1 Kbytes, then the logical address should have 

☒ 15 bits☐ 16 bits☐ 10 bits

☐ 14 bits

 **Will be reviewed**

34

Which **desk scheduling algorithm** does suffer from the **starvation problem**? 

☐ FIFO

☒ **Shortest-seek-time-first**

☐ C-SCAN

☐ SCAN

 **Will be reviewed**

35

What **hardware mechanism** is required in order to make **virtual memory systems efficient**? 

☐ EROM

☐ EEROM

☒ **TLB**

☐ DMA

 **Will be reviewed**

36

In the Banker's algorithm, the content of the **matrix Need** is 

- ☐ Allocation – Available
- ☐ Allocation – Max
- ☐ Max – Available
- ☒ **Max – Allocation**

 **Will be reviewed**

37

A virtual address is identical to a logical address. 

- ☒ **True**
- ☐ False

 **Will be reviewed**

38

A **disadvantage** of segmentation is that **segment tables** require a significant amount of space in memory 

- ☒ **True**
- ☐ False

 Will be reviewed

39

1. The following pseudo-code solves the critical section problem for two processes.

```
boolean flag[2];
flag[0]=flag[1]=false;
proc(int i) {
    while(true) {
        while(flag[(i+1) mod 2] == true);
        flag[i]= true;
        critical_section;
        flag[i] = false;    } }
```

☐ True

☒ False

 Will be reviewed

40

The following system of four processes with two resources:

If the availability vector is [2 3], the system above is deadlocked.



Current allocation matrix:	Current request matrix
P ₁ 1 3	P ₁ 1 2
P ₂ 4 1	P ₂ 4 3
P ₃ 1 2	P ₃ 1 7
P ₄ 2 0	P ₄ 5 1

☒ True

☐ False

✍ Will be reviewed

41

The following system of four processes with two resources:

If the availability vector is [2 4], the system above is deadlocked.



Current allocation matrix:	Current request m
P ₁ 1 3	P ₁ 1 2
P ₂ 4 1	P ₂ 4 3
P ₃ 1 2	P ₃ 1 7
P ₄ 2 0	P ₄ 5 1

☐ True

☒ False

✍ Will be reviewed

42

Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 2 sec of CPU, Job B is 8 secs, and Job C is 7 secs. Assume a time-slice of 1 sec. Given a FIFO scheduler, the turnaround time of job B is

☐ 11

☐ Others

☒ 10

☐ 8☐ 6☐ 4

 **Will be reviewed**

43

Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 2 sec of CPU, Job B is 8 secs, and Job C is 7 secs. Assume a time-slice of 1 sec. Given a FIFO scheduler, the average response time of the three jobs is 

☐ 6☐ 2☐ 11☐ Others☒ 4☐ 1

 **Will be reviewed**

44

Assume three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 2 sec of CPU, Job B is 8 secs, and Job C is 7 secs. Assume a time-slice of 1 sec. Given a RR scheduler, the average response time of the three jobs is 

☐ 2☒ 1☐ Others☐ 6☐ 4☐ 8

 **Will be reviewed**

45

A **binary semaphore** is a semaphore with **integer values** _____ 

☐ -1☒ 1☐ 2☐ -2

 **Will be reviewed**

46

The address generated by the CPU is referred to as _____ 

☒ **Logical address**☐ Physical address

- ☐ None of the mentioned
- ☐ Neither physical nor logical

 **Will be reviewed**

47

What is **compaction**? 

- ☐ a technique for overcoming internal fragmentation
- ☒ a technique for overcoming external fragmentation
- ☐ a paging technique
- ☐ a technique for overcoming fatal error

 **Will be reviewed**

48

_____ is generally **faster than** _____ and _____ 

- ☒ first fit, best fit, worst fit
- ☐ best fit, first fit, worst fit
- ☐ worst fit, best fit, first fit
- ☐ none of the mentioned

 **Will be reviewed**

49

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the **SCAN** algorithm? 

- ☐ 25, 50, 125, 150
- ☐ 125, 150, 25, 50
- ☐ 50, 25, 125, 150
- ☒ 125, 150, 50, 25
- ☐ 150, 125, 50, 25
- ☐ 125, 25, 150, 50

 **Will be reviewed**

50

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the **C-SCAN** algorithm? 

- ☐ 125, 150, 50, 25
- ☐ 50, 25, 125, 150
- ☐ 25, 50, 125, 150
- ☐ 125, 25, 150, 50
- ☒ 125, 150, 25, 50

☐ 150, 125, 50, 25

 **Will be reviewed**

51

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the **SSTF algorithm**? 

☐ 125, 150, 25, 50

☐ 150, 125, 50, 25

☐ 125, 150, 50, 25

☐ 25, 50, 125, 150

☐ 125, 25, 150, 50

☒ 50, 25, 125, 150

 **Will be reviewed**

52

Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the _____ 

☒ Ready state

☐ Terminated state

☐ Running state

☐ Suspended state

 **Will be reviewed**

53

In paged memory systems, if the page size is **increased**, then the **internal fragmentation** generally 

- ☐ becomes more
- ☐ has no effect
- ☐ remains constant
- ☒ becomes less

 **Will be reviewed**

54

A system is in a **safe** state **only** if there exists a _____ 

- ☐ safe allocation
- ☐ safe resource
- ☒ safe sequence
- ☐ all of the mentioned

 **Will be reviewed**

55

The **number** of processes completed **per unit time** is known as _____ 

- ☐ Efficiency
- ☒ **Throughput**
- ☐ Output
- ☐ Capacity

 **Will be reviewed**

56

Round robin scheduling falls under the category of _____ 

- ☐ Non-preemptive scheduling
- ☒ **Preemptive scheduling**
- ☐ All of the mentioned
- ☐ None of the mentioned

 **Will be reviewed**

57

Once a system is in an **unsafe state** **eventual** deadlock is **certain** 

- ☒ True

☐ False

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